

Business Problem Document

Capstone Project 2: Using Big Five Personality Traits to Identify Depression & Anxiety Risk

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Course: Intermediate AI & Data Science

Institution: AI Tech Institute

1. Executive Summary

This project develops a machine learning model to identify individuals at risk of depression and anxiety using personality traits from the Big Five Inventory (BFI). By leveraging research-validated relationships between personality and mental health, the model enables early screening and intervention opportunities.

Key Achievement: 95.7% F1-Score using a research-based multi-trait composite approach.

2. Problem Statement

2.1 The Challenge

Mental health conditions affect millions of people worldwide, yet early detection remains a significant challenge:

- **1 in 5** adults experience mental illness annually
- **60%** of those affected don't receive treatment
- **11 years** average delay from symptom onset to treatment

Early identification of at-risk individuals could enable proactive intervention and support.

2.2 Research Question

Can validated personality assessments predict mental health risk, enabling earlier identification of individuals who may benefit from support or intervention?

3. Solution Approach

3.1 Data Source

Big Five Inventory (BFI) - Real psychological survey data: - **Source:** Rdatasets Repository (academic data) - **Sample Size:** 2,436 respondents (after cleaning) - **Demographics:** 805 males, 1,631 females - **Data Type:** 100% real survey responses

3.2 Features

Six personality-based features: 1. **Neuroticism** - Anxiety, mood swings, emotional instability (N1-N5) 2. **Extraversion** - Social energy, positive emotions (E1-E5) 3. **Conscientiousness** - Self-discipline, organisation (C1-C5) 4. **Agreeableness** - Cooperation, trust (A1-A5) 5. **Openness** - Creativity, curiosity (O1-O5) 6. **Age** - Demographic factor

3.3 Methodology

Research-Based Multi-Trait Composite Score:

$$\text{Risk Score} = (\text{Neuroticism} \times 2) - \text{Extraversion} - (\text{Conscientiousness} \times 0.5)$$

Why This Approach: 1. Supported by multiple published research papers 2. Uses more available data than single-trait approaches 3. Captures the "High Neuroticism + Low Extraversion" pattern identified in research 4. Produces balanced class distribution using tertiles 5. Scientifically defensible methodology

3.4 Target Variable

Three-class classification using tertiles: - **Low Risk** - Bottom 33% of risk scores - **Medium Risk** - Middle 33% - **High Risk** - Top 33%

4. Model Performance

4.1 Results Summary

Model	Accuracy	F1-Score
Random Forest	95.7%	95.7%
Logistic Regression	95.5%	95.5%
Gradient Boosting	95.3%	95.3%
SVM	94.8%	94.8%
KNN	89.2%	89.1%

4.2 Gender Comparison

Population	Samples	F1-Score
All Combined	2,436	95.7%
Male Only	805	95.6%
Female Only	1,631	95.7%

Key Finding: Model performs consistently across both genders.

5. Key Findings

5.1 Top Predictors

1. **Neuroticism** - Strongest positive predictor of risk
2. **Extraversion** - Protective factor (lower = higher risk)
3. **Conscientiousness** - Protective factor

5.2 Research Alignment

- Results confirm Costa & McCrae (1992) findings
- "High Neuroticism + Low Extraversion" pattern validated
- Multi-trait approach outperforms single-trait models

6. Business Applications

6.1 Wellness Programs

- Early screening during wellness intake
- Risk stratification for resource allocation
- Personalised intervention recommendations

6.2 Research Applications

- Academic study of personality-mental health relationships
- Validation of established psychological theories
- Foundation for longitudinal studies

7. Ethical Considerations

7.1 Limitations

- This is a **screening tool**, NOT a clinical diagnostic
- Personality is one factor among many affecting mental health
- Model trained on specific population - generalisation requires validation
- Should complement, not replace, professional assessment

7.2 Responsible Use

- Informed consent required for any implementation
- Privacy and data protection essential
- Results should support, not stigmatise, individuals
- Professional oversight recommended for any real-world application

8. Future Improvements

1. **External Validation** - Test on independent datasets

2. **Longitudinal Tracking** - Follow up on predictions over time
 3. **Feature Expansion** - Integrate other wellness metrics
 4. **Accessibility** - Develop user-friendly screening application
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9. Conclusion

This project demonstrates that personality traits, particularly the combination of high Neuroticism and low Extraversion, can effectively identify individuals at higher risk of depression and anxiety. The research-based methodology achieves 95.7% accuracy and performs consistently across gender groups.

The model provides a foundation for early screening in wellness contexts, with appropriate ethical safeguards and professional oversight.
